

MADHUGHNEA SAI

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Education

Indiana University Bloomington

August 2023 – May 2025

M.S in Computer Science - Software Engineering, Distributed Systems, Adv Machine Learning

GPA: 3.7/4.0

Vellore Institute of Technology

July 2019 – June 2023

B.Tech in Computer Science and Engineering - Operating Systems, Distributed Systems, Networks and Security

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, HTML, CSS, C/C++, Java, SQL, Bash

Frameworks/Technologies: React, Angular, Node.js, Django, REST, jQuery, Agile, Spring Boot

Databases: MongoDB, PostgreSQL, MySQL

Industry Tools: [AWS](#) Developer Associate, Git, TensorFlow, JIRA, Docker, CI/CD, Firebase, Tableau, Jenkins, Postman

Experience

Full Stack Software Engineer, Y Stem and Chess

May 2025 - Present

- Migrated student activity portal from **Angular** to **React**, improving Lighthouse performance score from 68 to 90
- Deployed modular services on **Azure VMs** by custom domains & HTTPS via Terraform, reducing ops overhead by 20%
- Introduced **GitHub Actions** + **Docker multi-stage builds** to replace legacy Jenkins jobs, reducing deployment time and eliminating environment drift.

Software Developer intern, Central Indiana Corporate Partnership

July 2024 - December 2024

- Engineered a TensorFlow, Node.js **Microservice** to parse, extract key metadata from PDF job ads; increased system scalability, resilience by decoupling parsing logic from Sovren a third-party service
- Implemented a high-volume Selenium **data ingestion pipeline** with rotating VPN pools and CAPTCHA solvers, harvesting thousands of LinkedIn, career-site postings daily and cutting manual data entry 50%
- Leveraged **AWS (Lambda, S3)**; managed background tasks via Lambda to deal with variable web layouts, stored the scraped data in S3 for secure, scalable access; reduced bottlenecks by **35%**.
- Integrated a **React** Frontend with backend microservices via **RESTful APIs**, providing real-time job board updates.
- Established a **Jenkins CI/CD** pipeline with unit tests, Dockerized builds, shrinking release cycles by 30%.

Software Analyst intern, JP Morgan

June 2022 - July 2022

- Drove migration of over 200 critical tables, using JavaSpark, Hadoop & **SQL** to auto-detect and fix column drift, yielding 95%+ clean transfers across all business units.
- Diagnosed recurrent JavaSpark exceptions, instituted comprehensive **unit-test** coverage and technical docs, lowering pipeline failure frequency by 20% and shrinking outage durations.
- Streamlined Control-M scheduling for 20 daily migration tasks and synced with SRE via ServiceNow, cutting escalation time 25% and ensuring smooth ops.

Projects

Agentic AI for Backend Operations - Hackathon Winner | Python, Langchain, Llama, NLP, AWS

Mar 2025

- Developed an LLM-powered AI agent that interprets natural language commands for CRUD operations on a key-value store—boosting backend efficiency by 30% and cutting processing errors by 25% with a slot-based extraction mechanism.
- Integrated a Gradio interface for real-time tracking and auditing feature to track changes

Real-time Market Prediction from Twitter | Python, Tweepy, GPT-3, LangChain, AWS

Jan 2025 - May 2025

- Developed a real-time data pipeline using Tweepy to gather financial tweets, applying noise filtering and multilingual support, then integrated GPT-3 via LangChain to boost sentiment classification accuracy by 25%.
- Developed a visualization dashboard to display sentiment trends, enabling 35% faster market trend predictions.

Optimization of Legacy Data: refactoring via MongoDB & AWS S3

July 2024 - Dec 2024

- **Decoupled dependent collections**, redesigned MongoDB data architecture for a **50%** reduction in per-document size.
- Integrated S3 data layer to Node.js backend, with legacy workflow compatibility; dropping data service **latency** by **30%**

Adv Secure Banking Platform

Jan 2024 - May 2024

- Delivered a secure full-stack banking solution with Angular front end, Spring Boot services, **PostgreSQL** DB in Docker deployments with role-based access control and stateless JWT auth, reducing unauthorized access by 50%
- Integrated Google Sign-Up (Firebase) plus advanced security checks, streamlining onboarding and delivered through SCRUM, test-driven sprints.